Wired 02 | Provisioning a Campus Fabric

Overview

In this lab, you'll onboard your switch using CloudVision Studios, adding a new campus leaf switch to an existing Campus Fabric. We'll take a switch from out-of-the-box Zero-Touch Provisioning (ZTP) mode to a fully provisioned member of the fabric — without the use of CLI or a console cable.

What you'll learn:

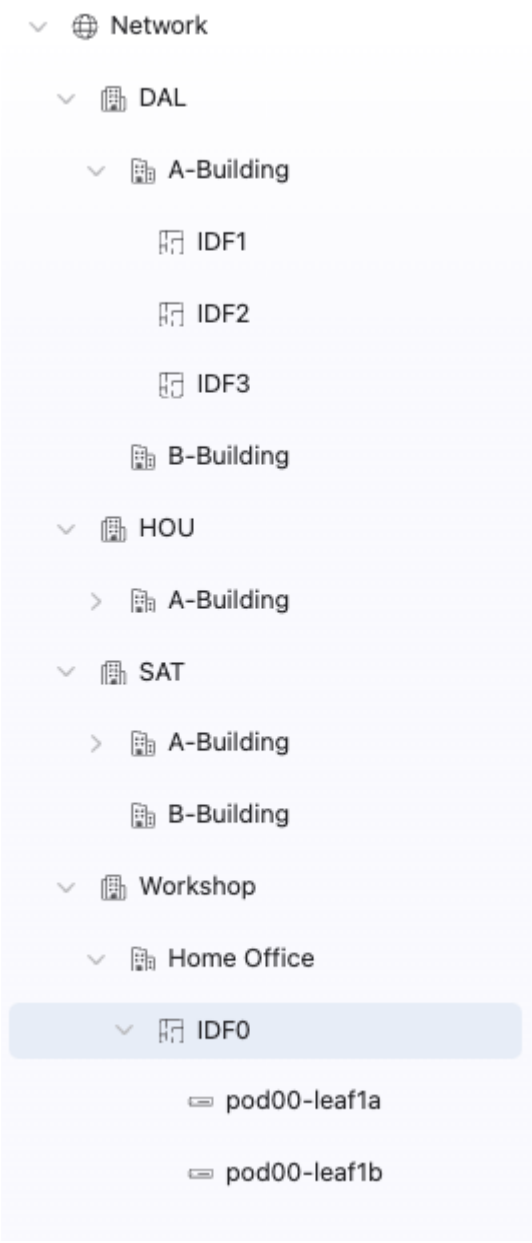
- Onboard a new switch using CloudVision Studios Quick Actions
- Navigate the Campus Fabric hierarchy (Campus, Pod, Access Pod)
- Configure spine uplinks for external devices
- Review and submit a workspace for change control
- Execute and monitor a change control

Your environment has been pre-configured with a sample Campus to assist with these new concepts. Studios uses flexible constructs to describe your campus footprint:

- **Campus Fabric**: Your campus `Workshop` has been created — this is equivalent to a site or region.
- **Campus Pod**: Your pod `Home Office` could be a building or MDF.
- **Access Pod**: Your access pods `IDF1` and `IDF2` could be a floor or IDF.

Show me an example!

CloudVision Network Hierarchy gives a better visual representation of how this might look in a real environment.



Account Assignments

You and and a fellow student will work separately to onboard your switches to you assigned IDFs. Student-1 is assigned to IDF1 and Student-2 is assigned to IDF2. We've assigned two accounts per pod. Use your pod number along with this account format to determine your user account. example: tola.atd `1` + `pod##` @gmail.com & tola.atd `2` + `pod##` @gmail.com

Open Launchpad | **Login Help** | **Lab Topology**

CloudVision Login

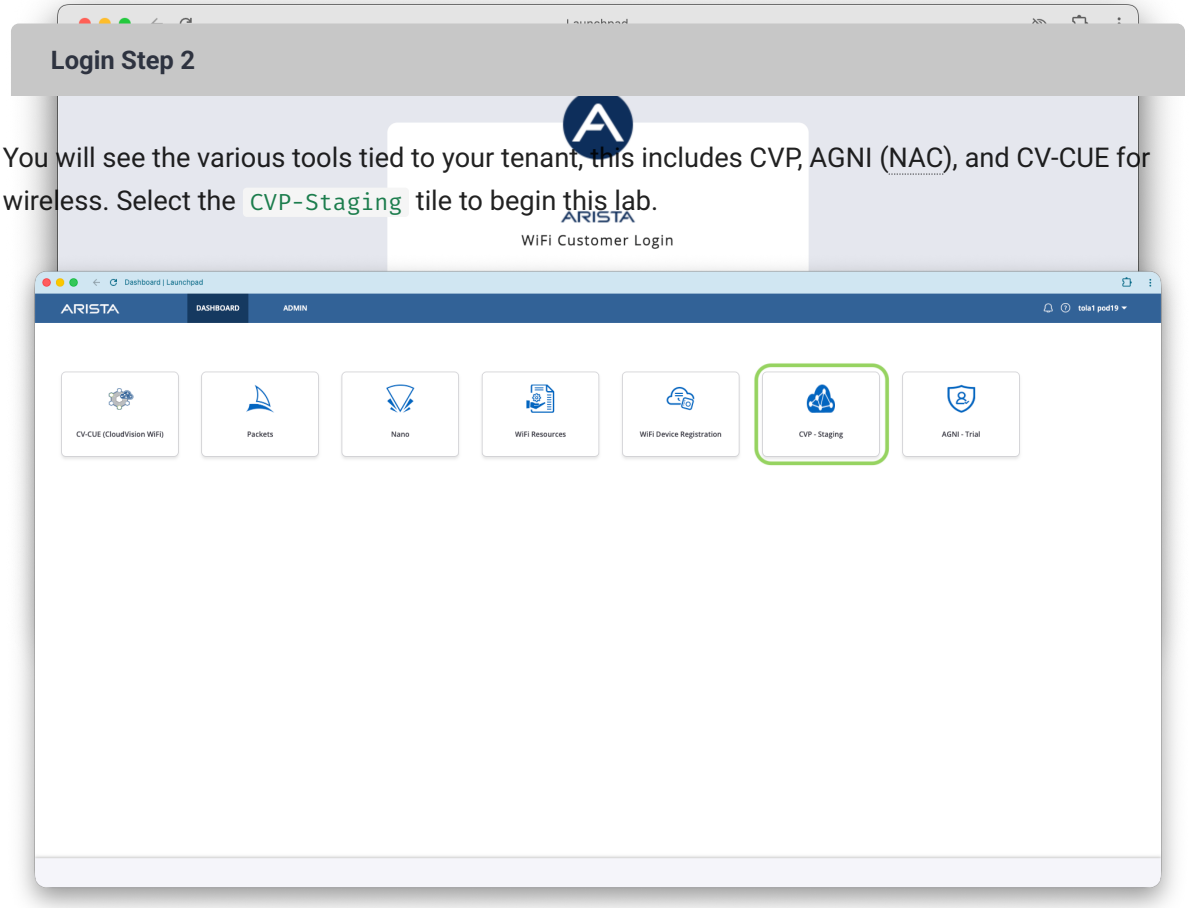
If you're not already logged into CloudVision (CVaaS), navigate to the Arista Launchpad for your lab.

Open Launchpad

Login Step 1

Select the `Log in with Launchpad` button and provide your assigned lab#+pod## email address and password:

Login Step 2

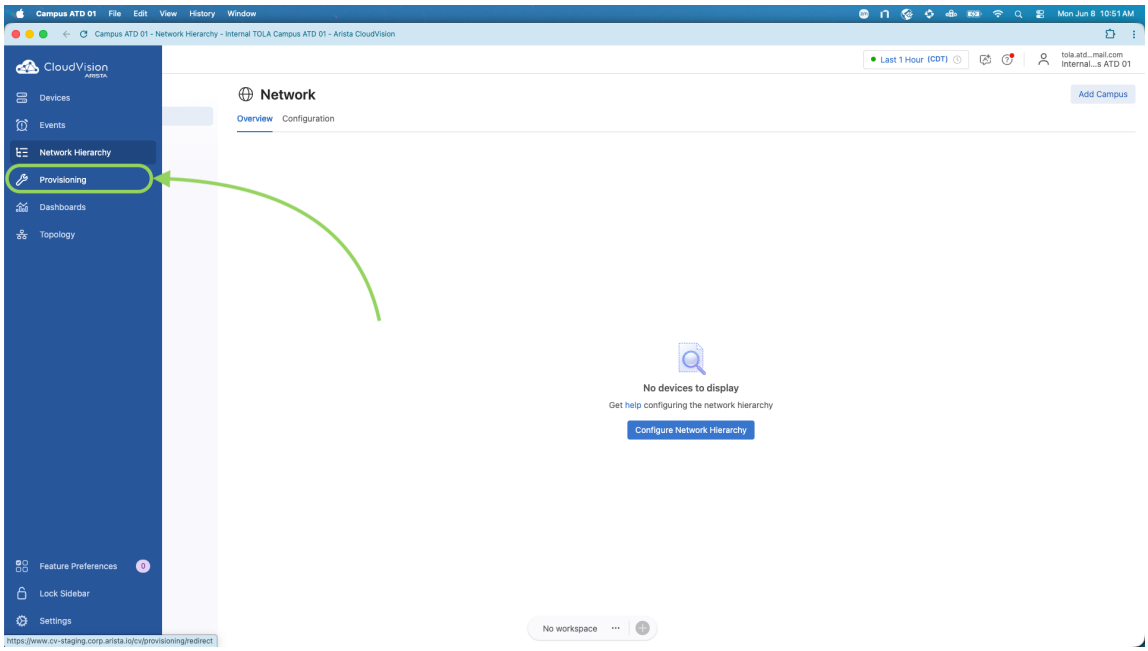


You will see the various tools tied to your tenant, this includes CVP, AGNI (NAC), and CV-CUE for wireless. Select the `CVP-Staging` tile to begin this lab.

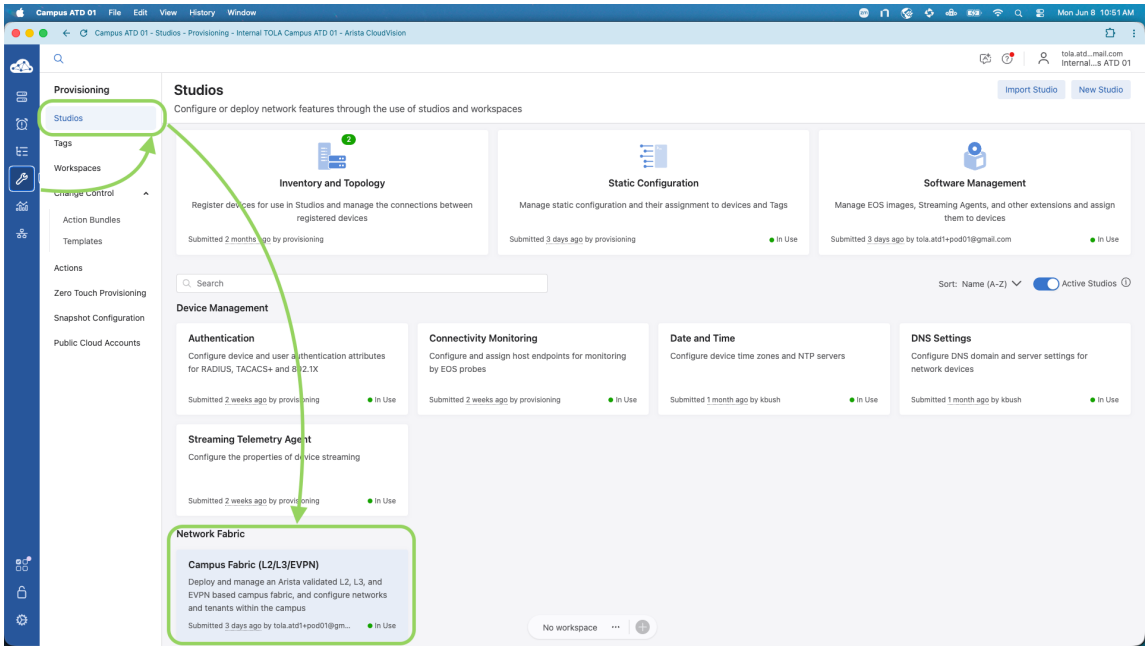
Onboarding Devices

Before you begin, we've assigned you and a fellow student to a `Pod`, and each `Pod` is assigned to a separate CloudVision Organization: TOLA-ATD-(01-12). We are using a bootstrap server to distribute tokens and separate the switches in to their assigned CVAAS Instance.

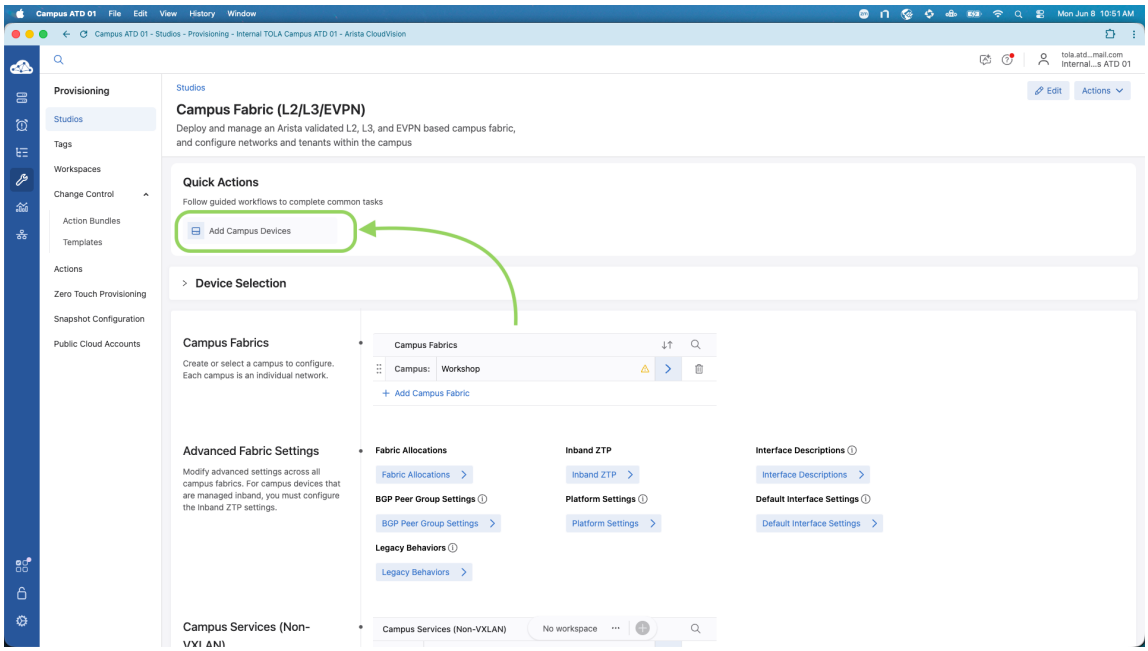
1. To begin onboarding your switches, navigate to the 🔧 `Provisioning` icon on the left.



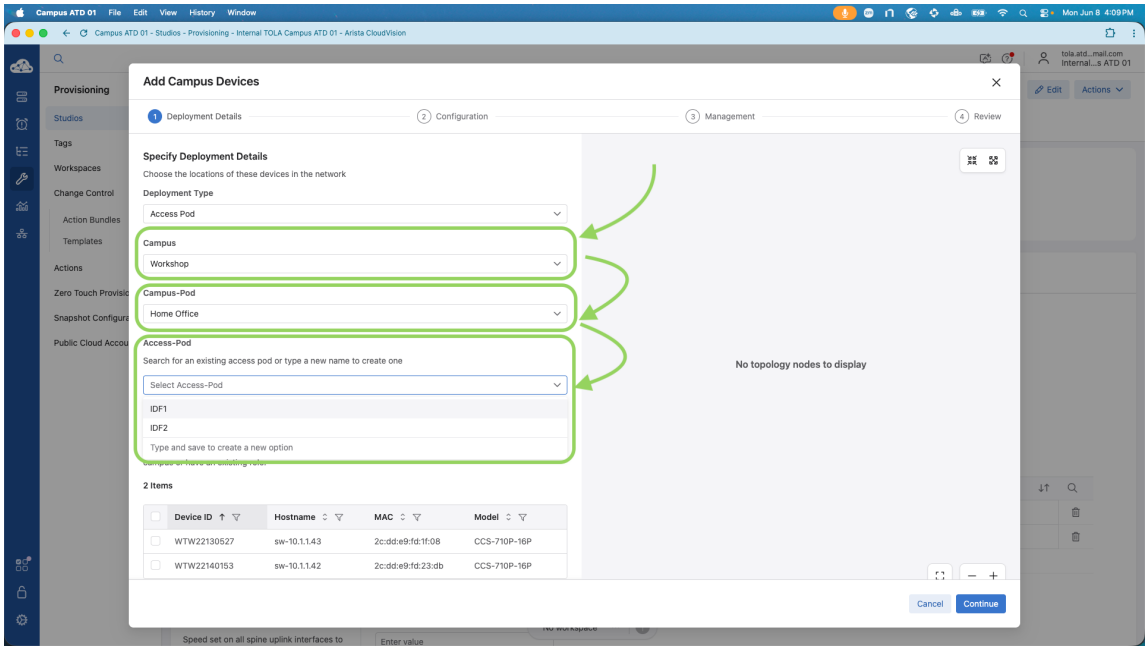
2. Navigate to `Studios` and select `Campus Fabric (L2/L3/EVPN)`. You can toggle `Active Studios` as needed to filter down the options.



3. Select `Add Campus Devices` under the Quick Actions menu



4. Select `Campus:Workshop`, `Campus-pod: Home-Office`, and `Access-Pod`, then select `IDF1` for Student 1 or `IDF2` for Student 2.



5. Use the table below determine your switches `Device ID` associated with your assigned `Hostname`

What's my hostname?

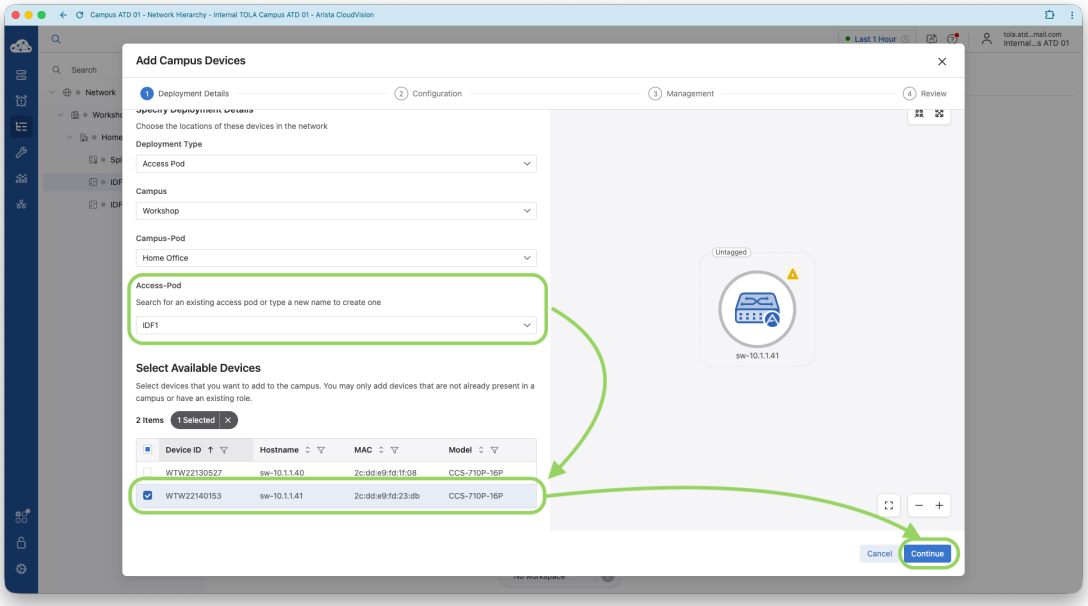
The hostname includes pod+student identification. Student 1 will choose `leaf1a` and Student 2 will choose `leaf1b` for Pods 1-12.

Switch Names

Hostname	Device ID	Hostname	Device ID
pod01-leaf1a	WTW22140153	pod07-leaf1a	WTW22130459
pod01-leaf1b	WTW22130527	pod07-leaf1b	WTW22140623
pod02-leaf1a	WTW22130402	pod08-leaf1a	WTW22130479
pod02-leaf1b	WTW22150107	pod08-leaf1b	WTW22130554
pod03-leaf1a	WTW22130409	pod09-leaf1a	WTW22140124
pod03-leaf1b	WTW22130472	pod09-leaf1b	WTW22130439
pod04-leaf1a	WTW22130476	pod10-leaf1a	WTW22130436
pod04-leaf1b	WTW22130478	pod10-leaf1b	WTW22130545
pod05-leaf1a	WTW22130506	pod11-leaf1a	WTW22130549
pod05-leaf1b	WTW22140213	pod11-leaf1b	WTW22130529
pod06-leaf1a	WTW22140174	pod12-leaf1a	WTW22130440
pod06-leaf1b	WTW22130505	pod12-leaf1b	WTW22130532

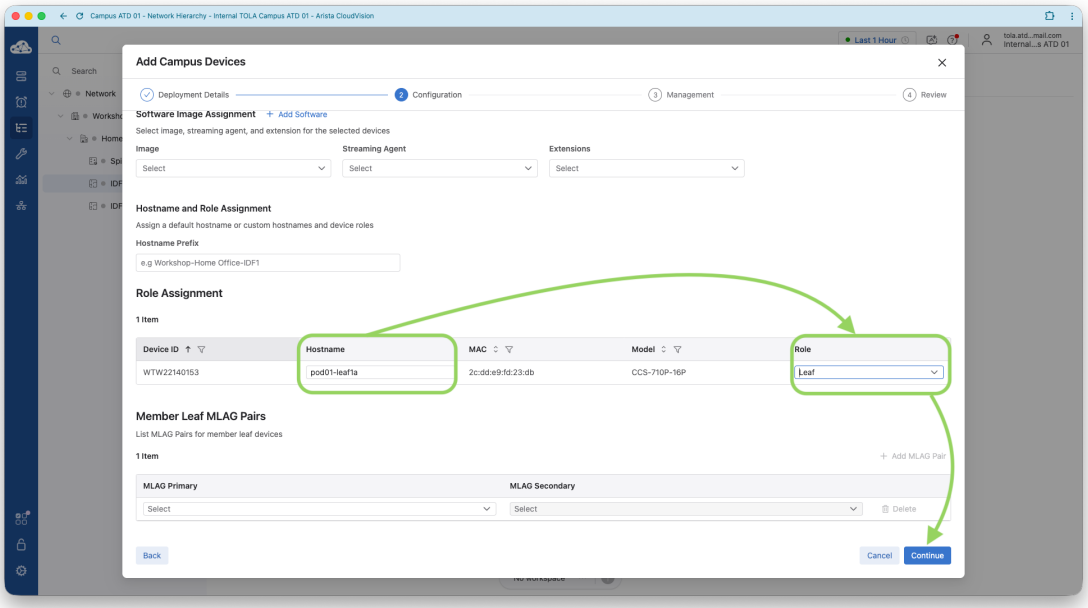
Step 1

`Continue`.



Step 2

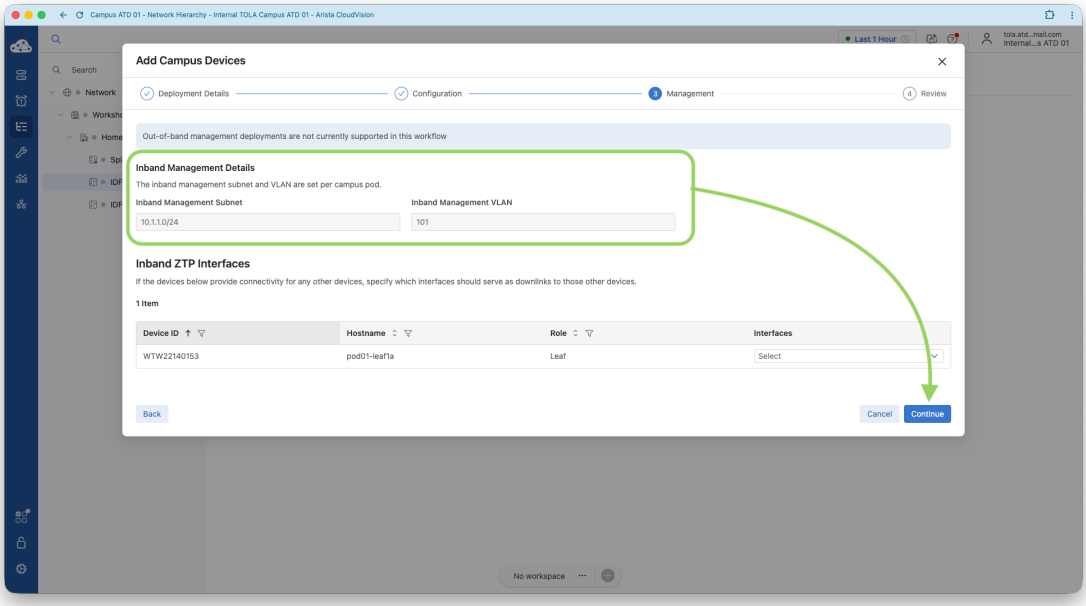
Select the `Hostname` field and type the name your switch. Next select the role as `Leaf` and select `Continue`.



Step 3

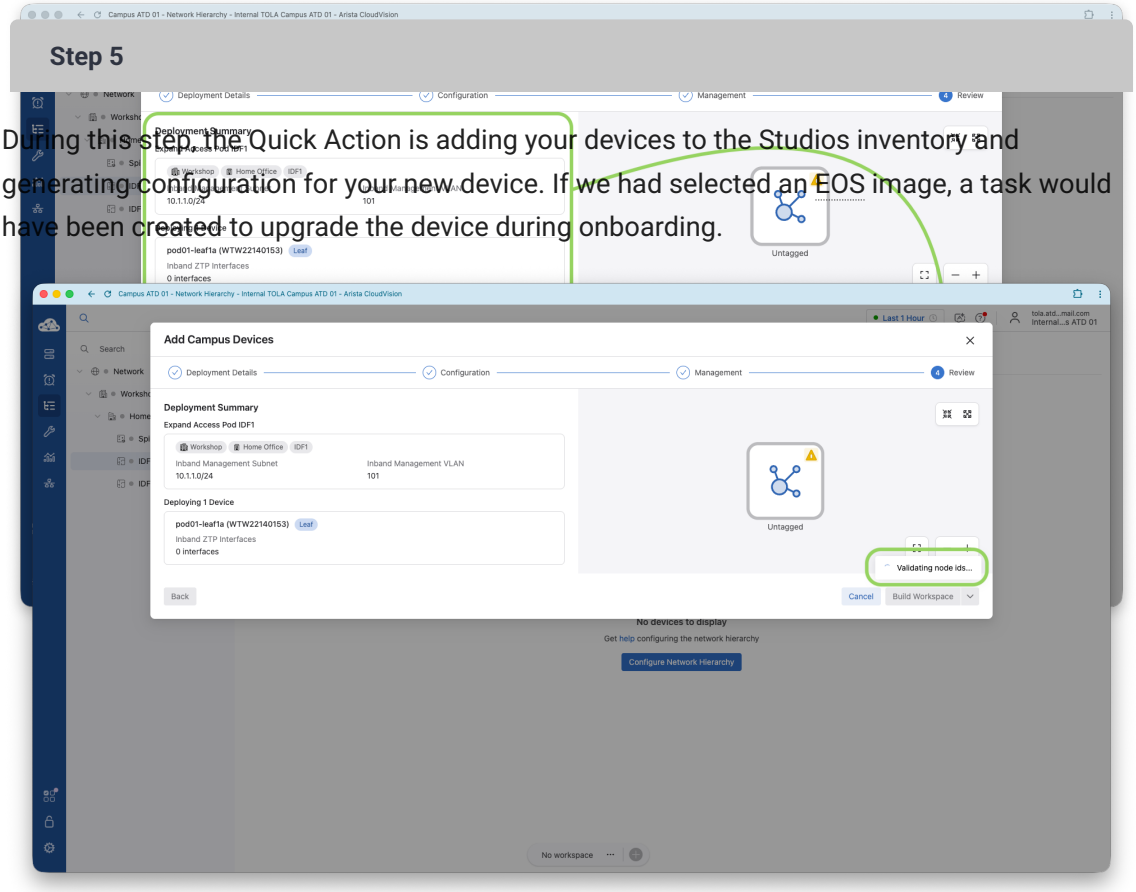
Validate your `Inband Management Subnet` and `Inband Management VLAN` match your assigned pod number. This was pre-configured in Studios.

- **Subnet**: `10.1.#.0/24` where `#` is the pod number
- **VLAN**: `1##` where `#` is the pod number



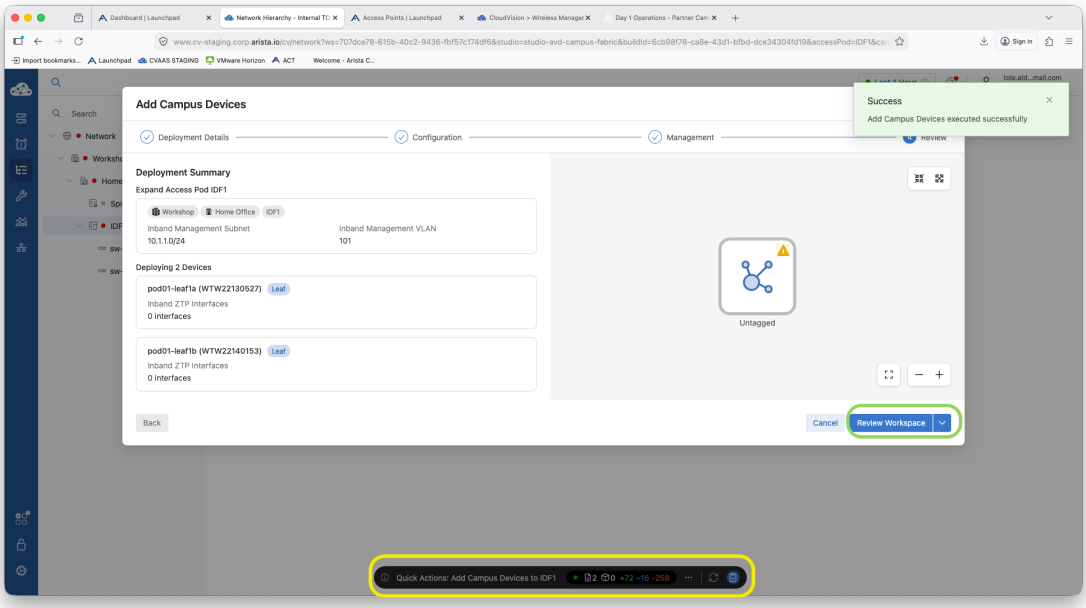
Step 4

This is a review of what you have configured to onboard your device. Select `Build Workspace` when you are ready.



During this step, the Quick Action is adding your devices to the Studios inventory and generating configuration for your new device. If we had selected an EOS image, a task would have been created to upgrade the device during onboarding.

6. The workflow is completing several steps to onboard the device, and automatically created a new Workspace noted at the bottom of your screen. Select `Review Workspace` to explore.



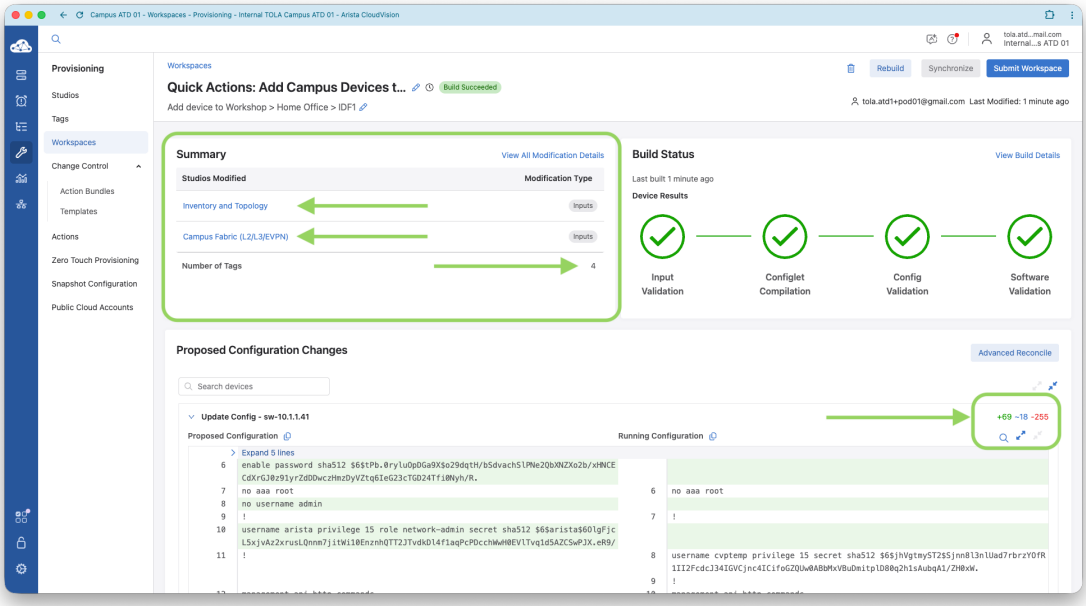
7. In the workspace, note the top-level `Summary` box. There are several studios modified:

   a. `Inventory and Topology`: Devices selected are added to the Campus inventory.

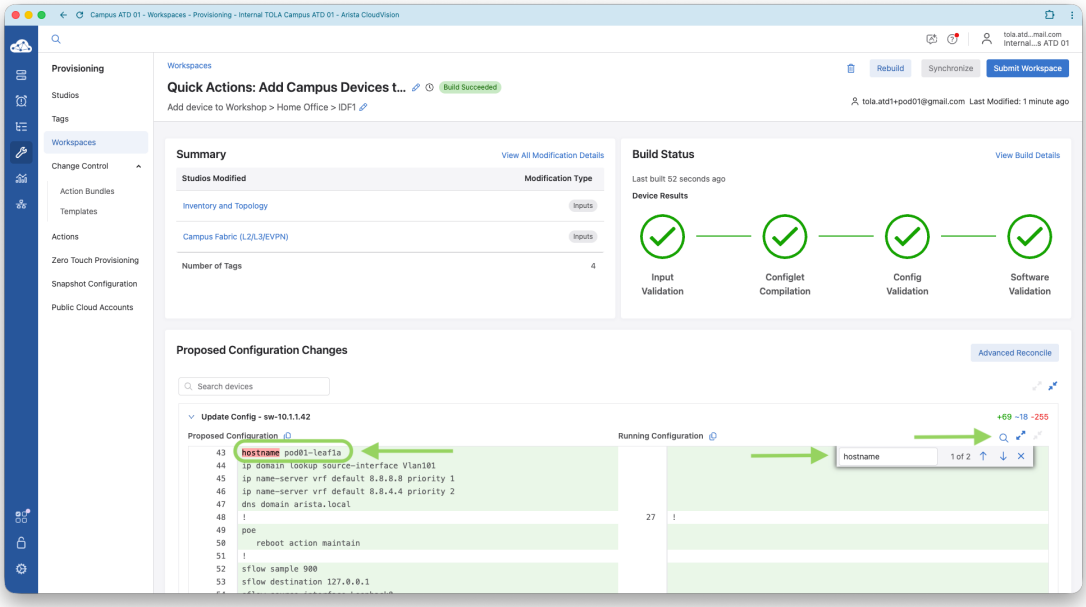
   b. `Campus Fabric (L2/L3/EVPN)`: Devices were added to their respective Campus Access Pod `IDF1` and will inherit configuration from that part of the campus.

   c. Note the number of Tags

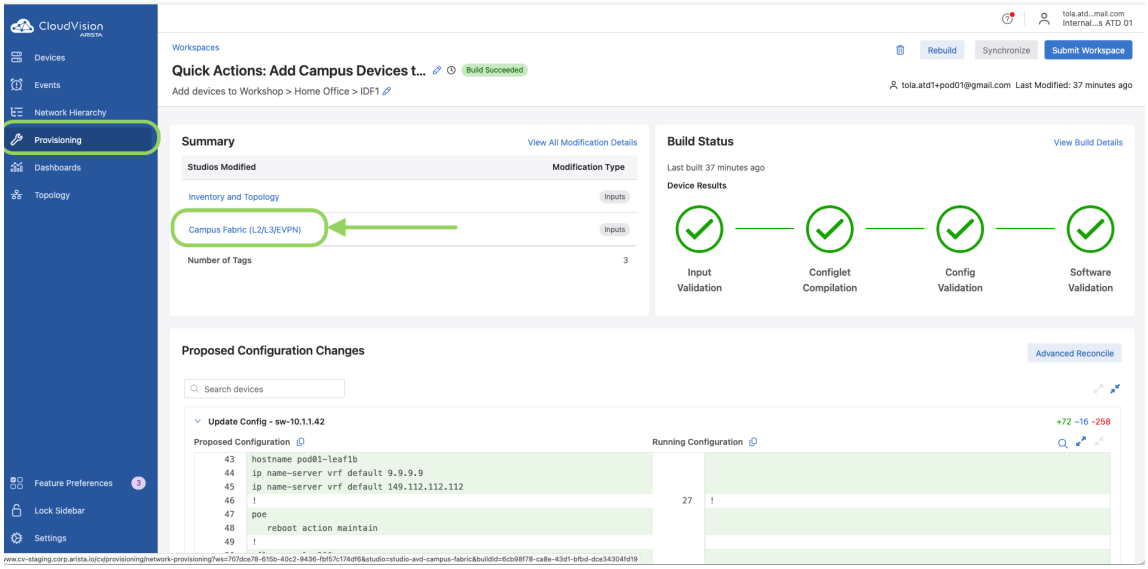
   d. Review the number of lines being added (green), modified (blue), and deleted (red) in the bottom right.



8. Next let's verify the hostname has changed in the `Proposed Configuration` by selecting the `Search` function and typing the `hostname`



9. Do **NOT** select `Submit Workspace`. Navigate back to the `Studios` by selecting `Campus Fabric (L2/L3/EVPN)` or going back to 🔧 `Provisioning`.



Did you submit your Workspace?

It's okay since no configuration change has been made. You will need to approve a Change Control later to configure the switch. Also, a new workspace will be automatically created when you begin making your next change.

Configuring Spine Uplinks

1. Next, we'll configure the uplinks in the `Campus Fabric (L2/L3/EVPN)` studio.

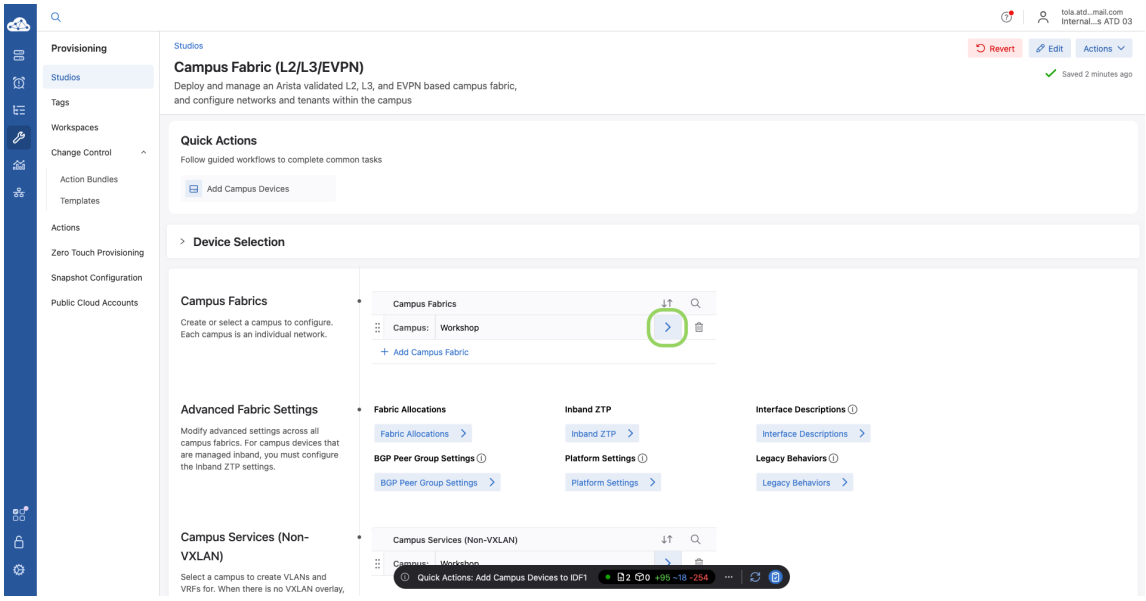
Spine Configuration

Normally, you wouldn't have to perform this step; however, since the spines are part of the Instructor's Fabric, we will need to manually configure the spine uplinks.

2. Next, we'll drill down to configure the spine uplinks.

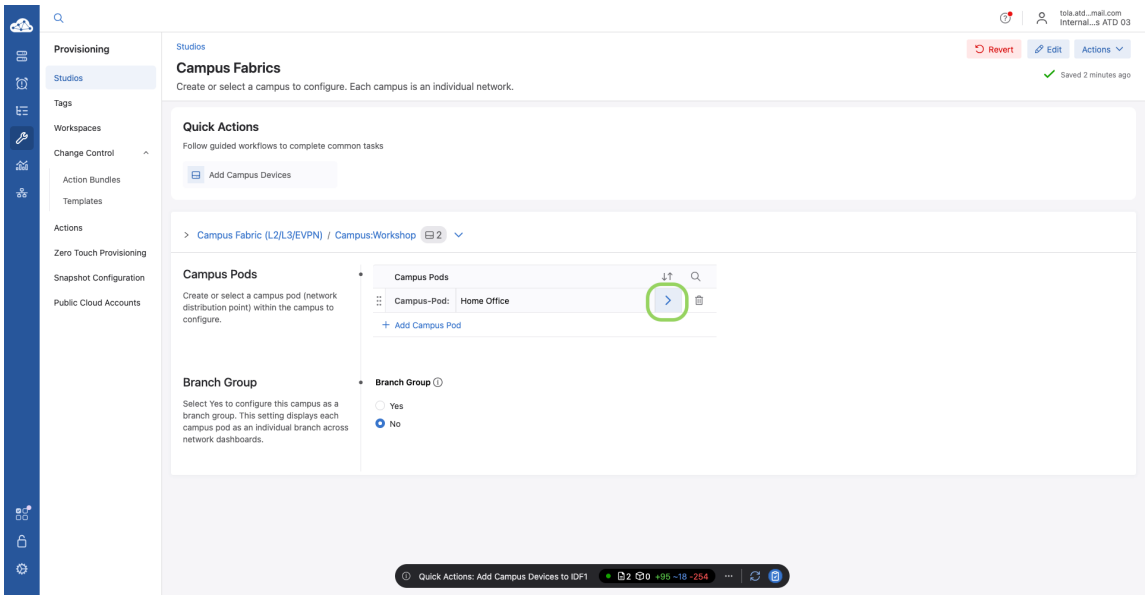
Step 1 - Spine Links

Select the blue arrow by `Campus: Workshop`



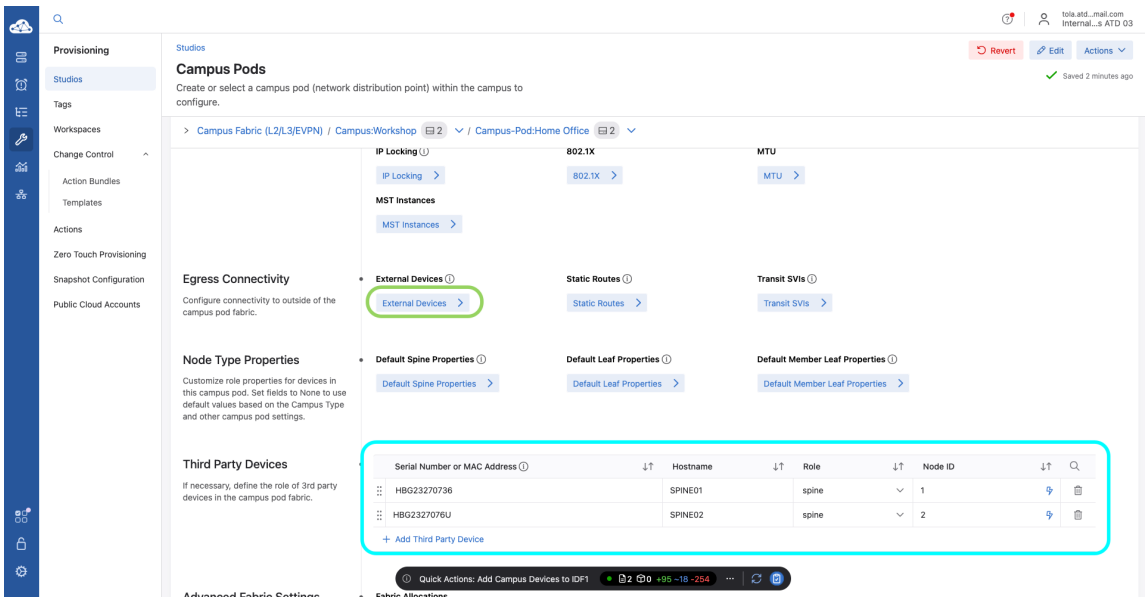
Step 2 - Spine Links

Select `CampusPod: Home Office`



Step 3 - Spine Links

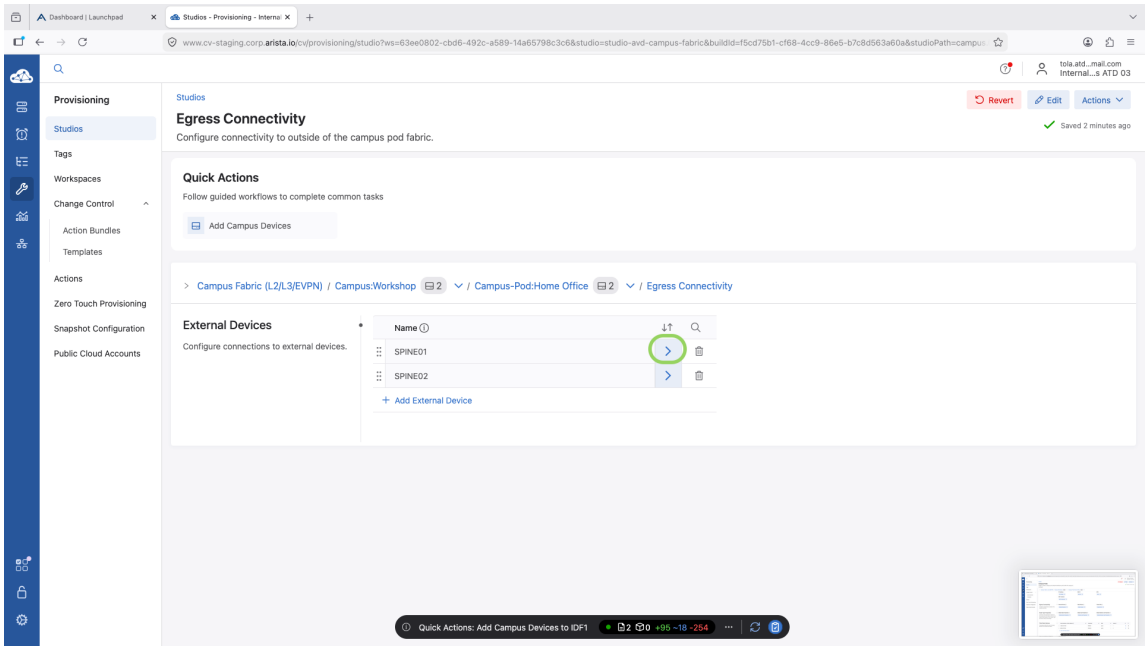
Scroll down to `Egress Connectivity` and select `External Devices`. Note the spines are configured as 3rd party devices below.



3. Now let's configure your switch uplinks for both Spines. Each switch has uplinks to both SPINE01 and SPINE02 on Ethernet Interfaces 15 and 16.

Step 1 - Select SPINE01

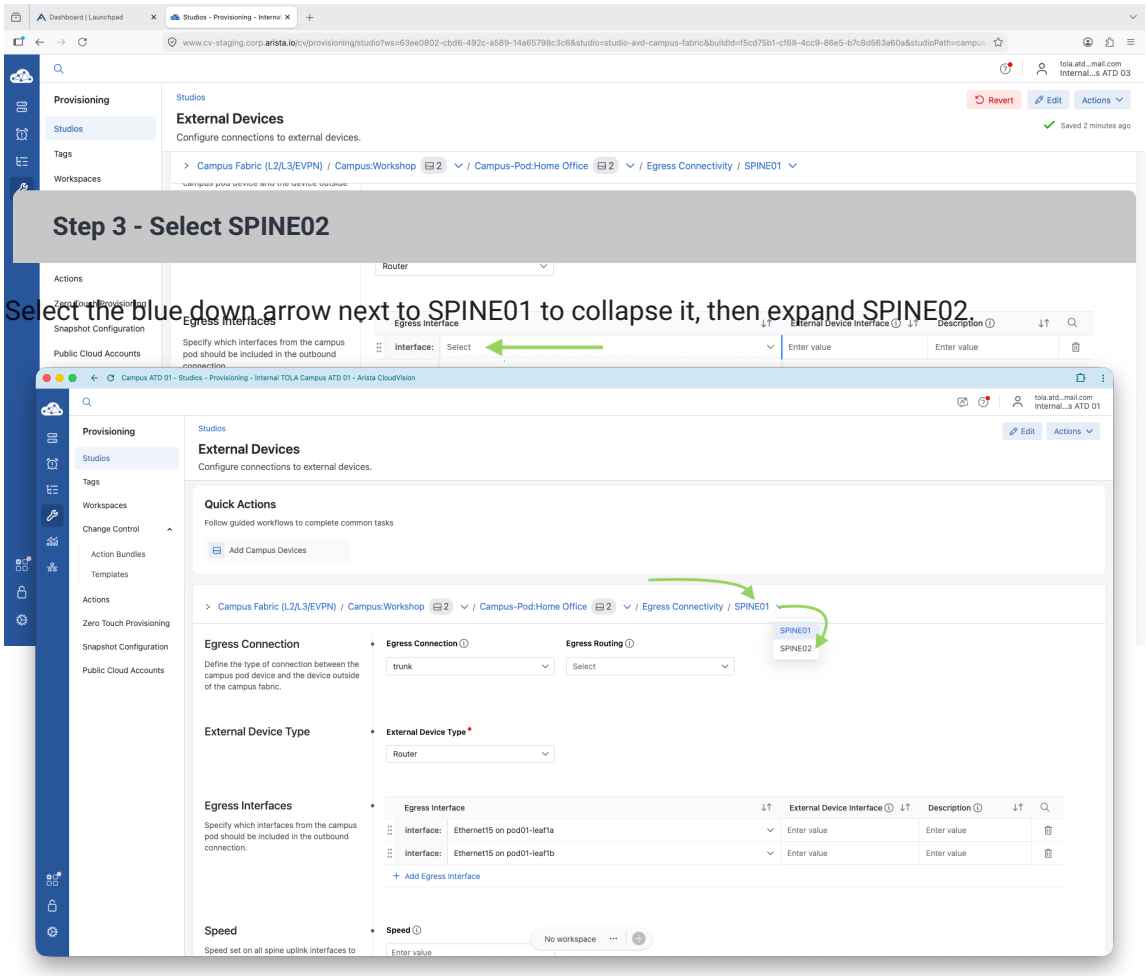
Select the blue arrow next to SPINE01.



Step 2 - Configure SPINE01

Scroll down to `Egress Interfaces`, then select the blank space next to `Interface` and type `15` in the `Interface:` field. Select the interface matching your student assignment:

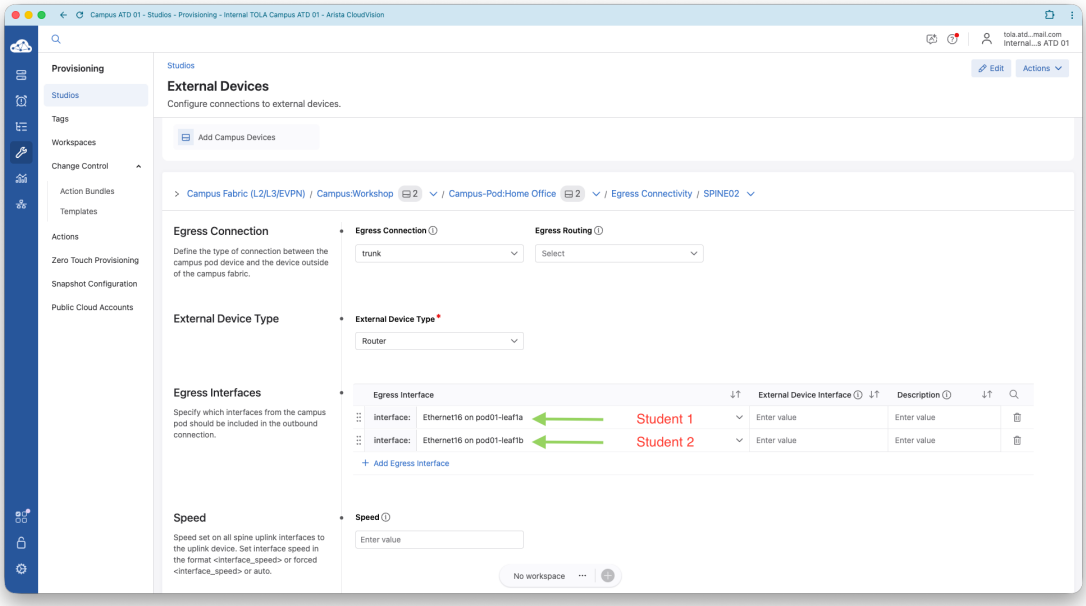
Student	Interface
Student 1	`Ethernet 15 on podXX-leafa`
Student 2	`Ethernet 15 on podXX-leafb`



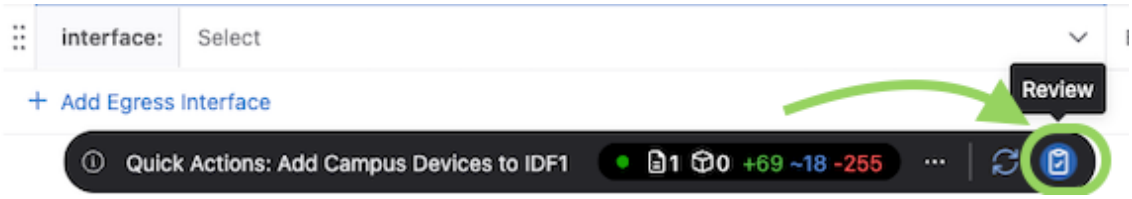
Step 4 - Configure SPINE02

Repeat the same process for interface 16. Select the interface matching your student assignment:

Student	Interface
Student 1	`Ethernet 16 on podXX-leafa`
Student 2	`Ethernet 16 on podXX-leafb`

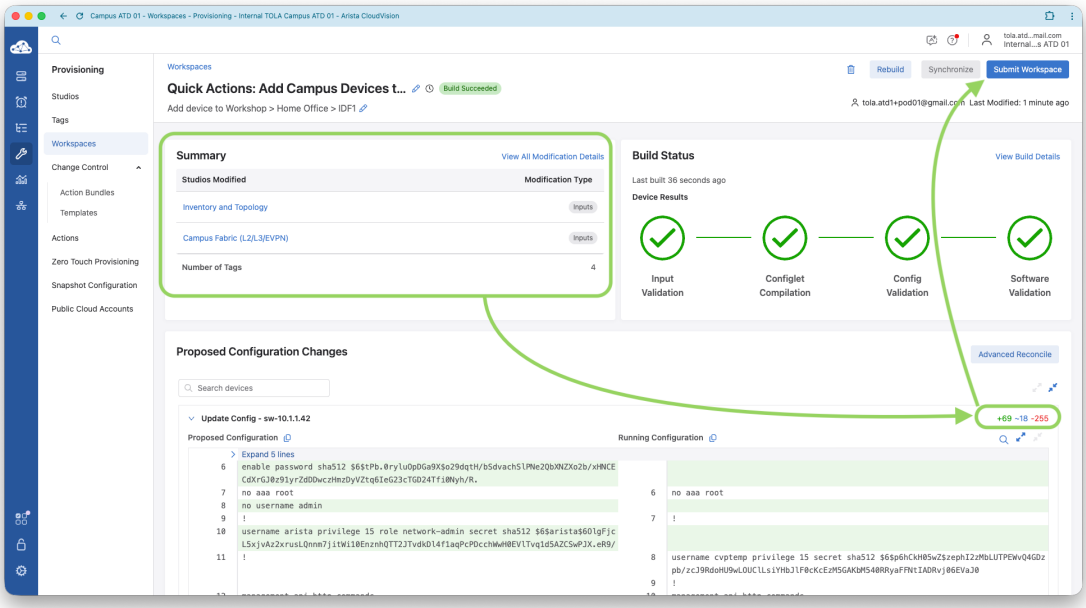


4. Let's now review our workspace and the changes we've made. Select `Review Workspace` on the Workspace Island.



5. You can review the configuration changes to the devices.

> **The Summary**
>
> You can see there are 2 Studios and tags being used. We've also pre-configured several other Studios (Authentication, DNS, Date/Time, Streaming Telemetry) for the base configuration of the switch instead of using a Static configuration.

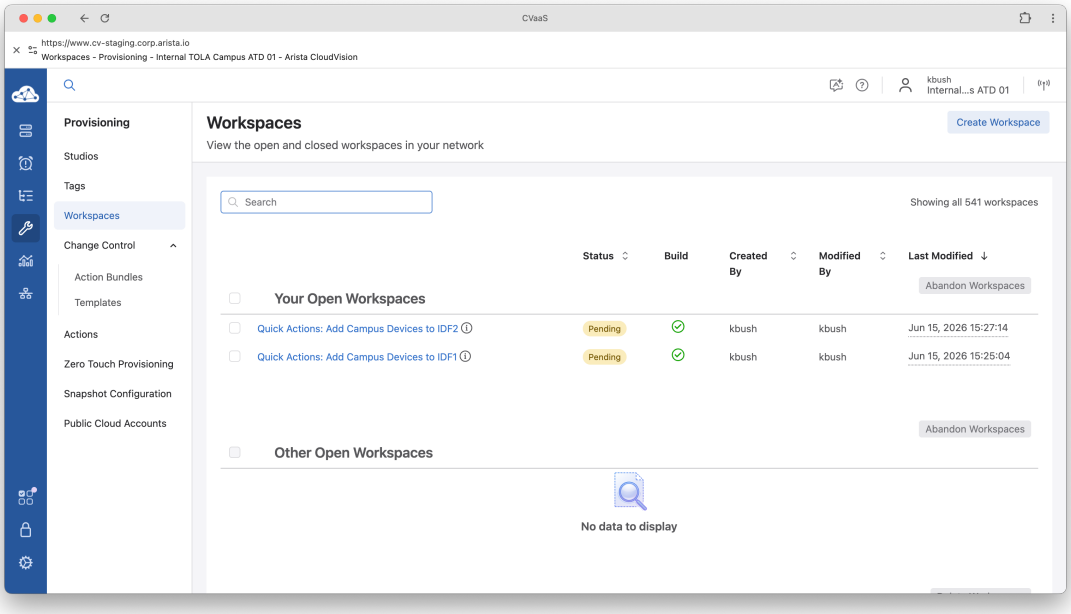


Submitting a Workspace

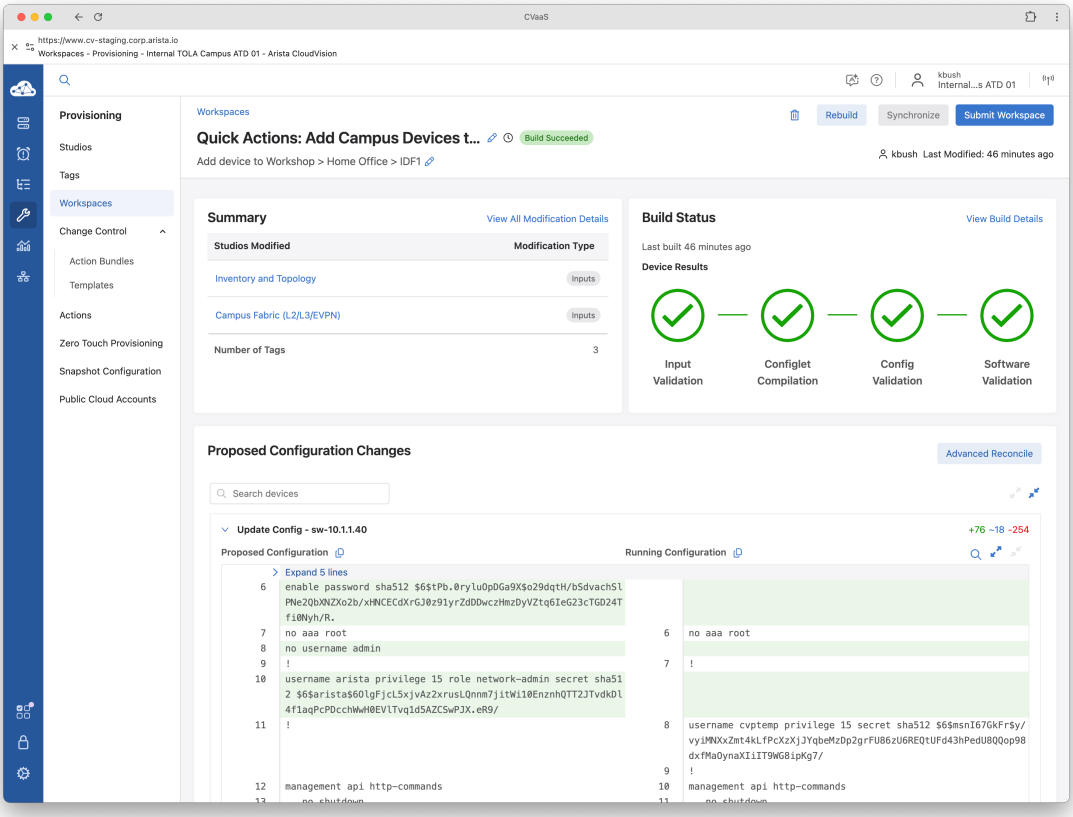
Up to this point, you have been making changes in parallel, each in their own workspace. Before these changes take effect, each workspace must be submitted — but only one can go first. When the second student submits, CloudVision will detect that the mainline has changed and require a synchronization — similar to pulling the latest changes before merging a branch in Git.

> **Workspace Synchronization**
>
> A workspace in CloudVision is like a feature branch. While you work, others may submit changes to the mainline. Before you can submit your own workspace, you need to synchronize — pulling in the latest mainline changes and rebuilding your configuration on top of them.

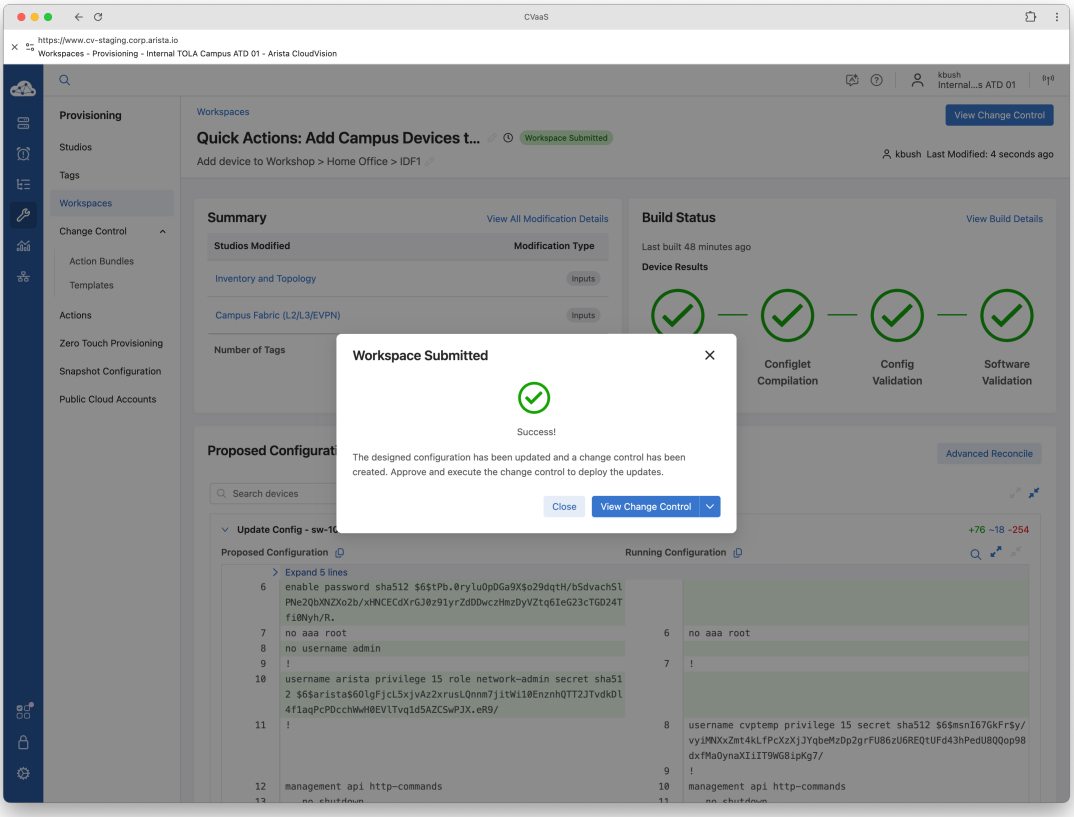
1. Select the `Workspaces` tab. You should see both workspaces — one for each student — adding devices to their respective IDFs.



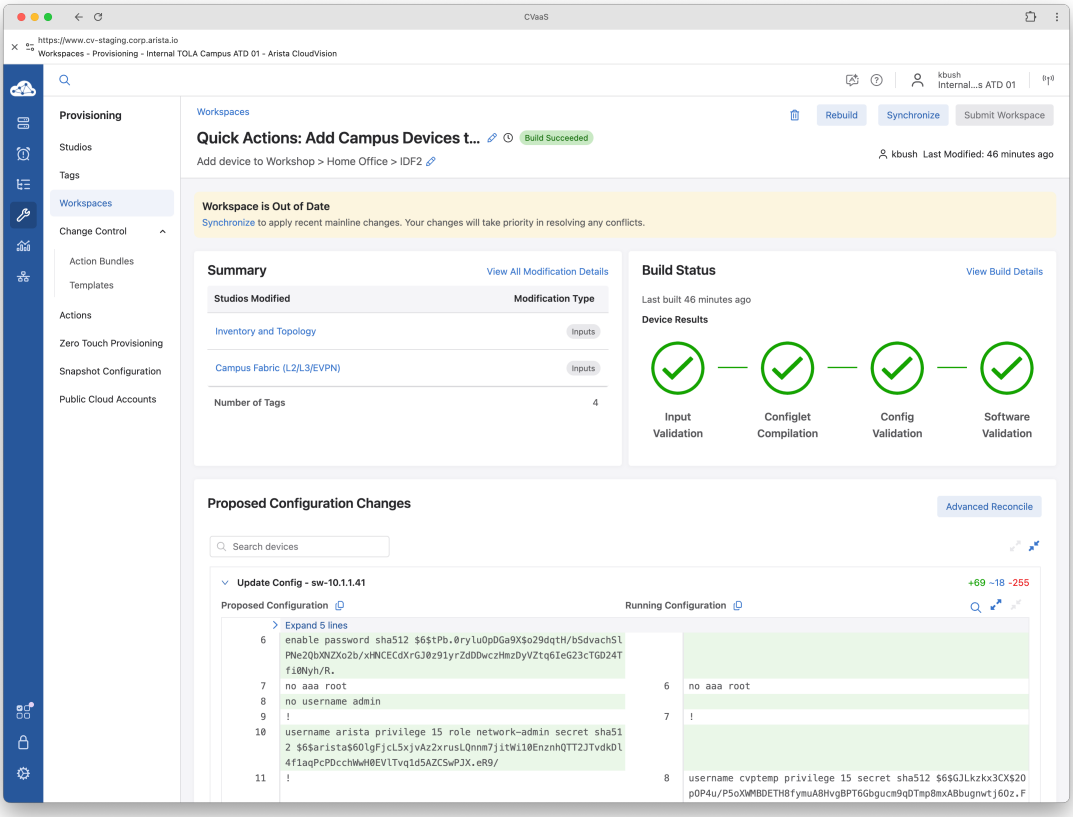
2. Select your workspace and notice the `Synchronize` button is greyed out. This is expected — the mainline hasn't changed yet.



3. **Student 1** should submit their workspace first by selecting `Submit Workspace`.



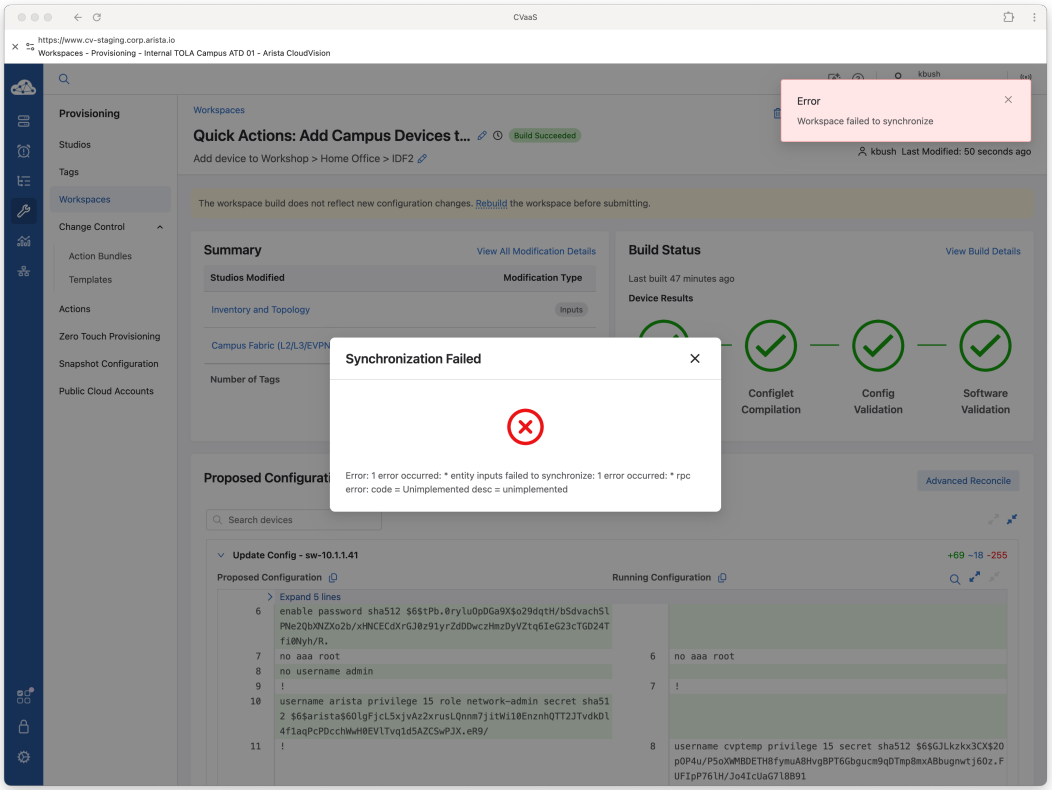
4. **Student 2** will now see a banner indicating `Workspace is Out of Date`. This is because Student 1's submission updated the mainline.

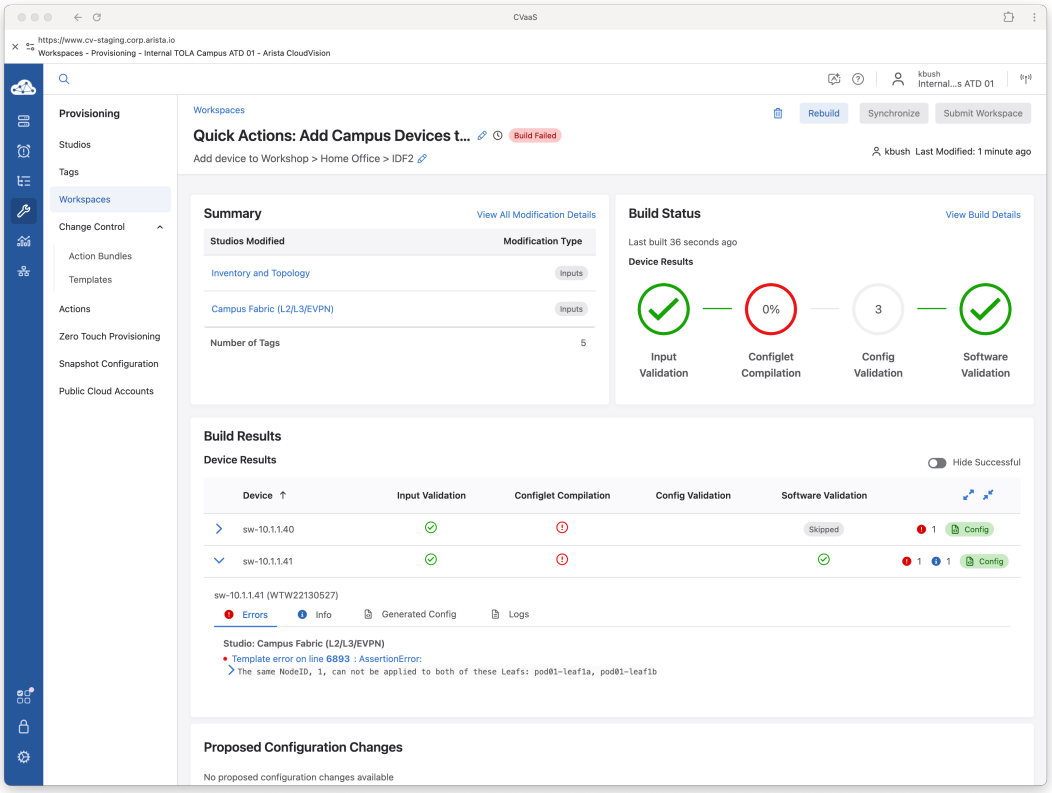


5. To resolve this, select `Synchronize`. CloudVision will rebuild your workspace on top of the latest mainline changes.

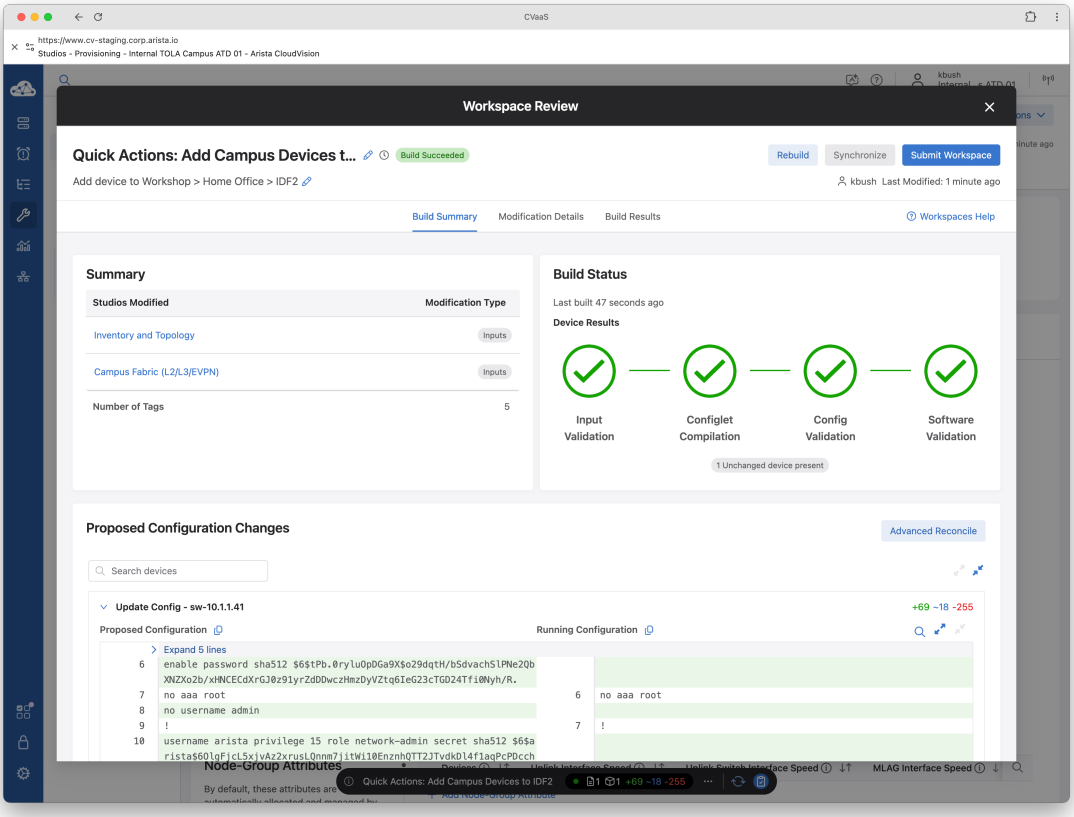
Known Issue: Synchronization Errors

You may encounter errors where Node IDs fail to recalculate during synchronization. If you see the errors below, don't worry — we'll resolve them together.

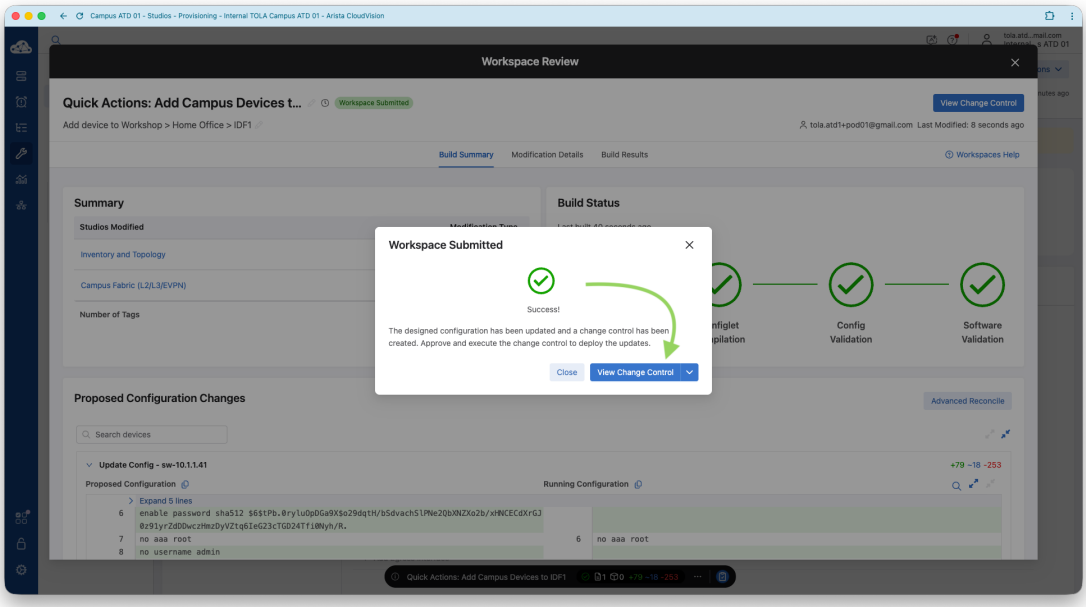




6. Once synchronized, **Student 2** can select `Submit Workspace` to generate their Change Control.

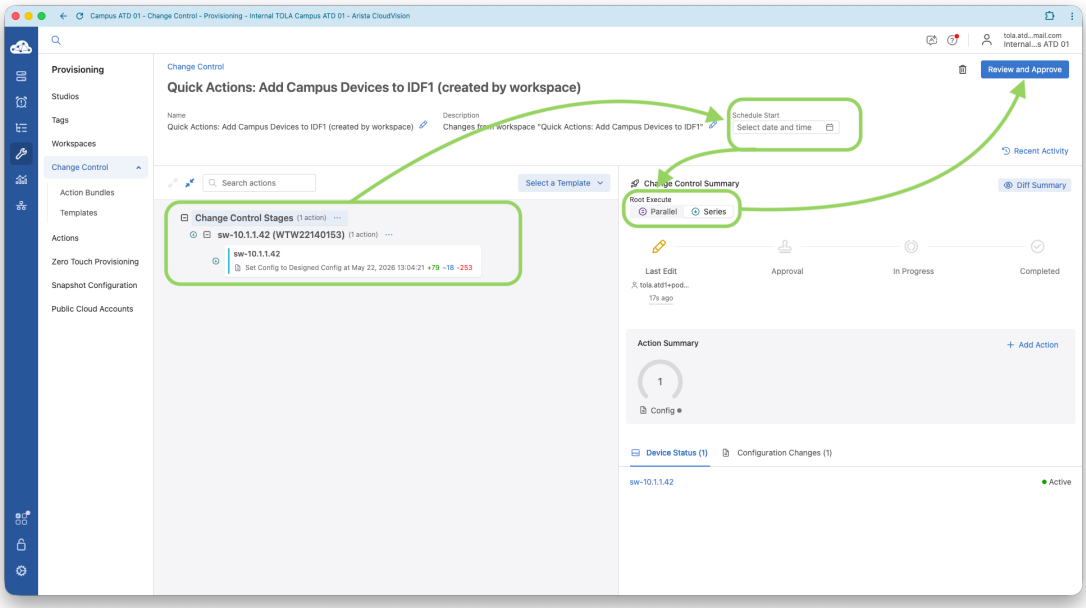


7. Select `View Change Control` — let's explore executing a change control in the next section.



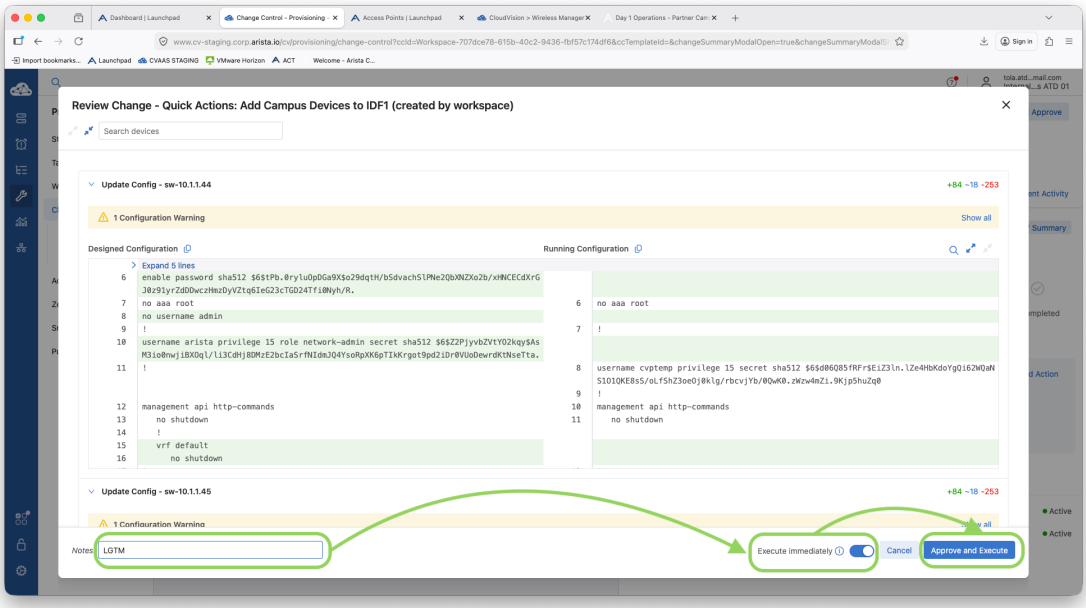
Executing a Change Control

1. Within the Change Control, you can review the configurations as we just did in the workspace. This encourages review of changes prior to execution.



2. When you are ready, select `Review and Approve` at the top right, toggle `Execute Immediately`, and select `Approve and Execute`.

> **I approved my own change?!**
>
> Yes, for this workshop you can approve your own change. In `Settings > General Settings`, the toggle for `Non-Author Change Control Review` can be enabled to enforce a change control review from a "non-author".



3. Let's explore what is going on during execution of the change control. While this is happening, feel free to select `Logs` near the top right to watch what's happening.

CC Start

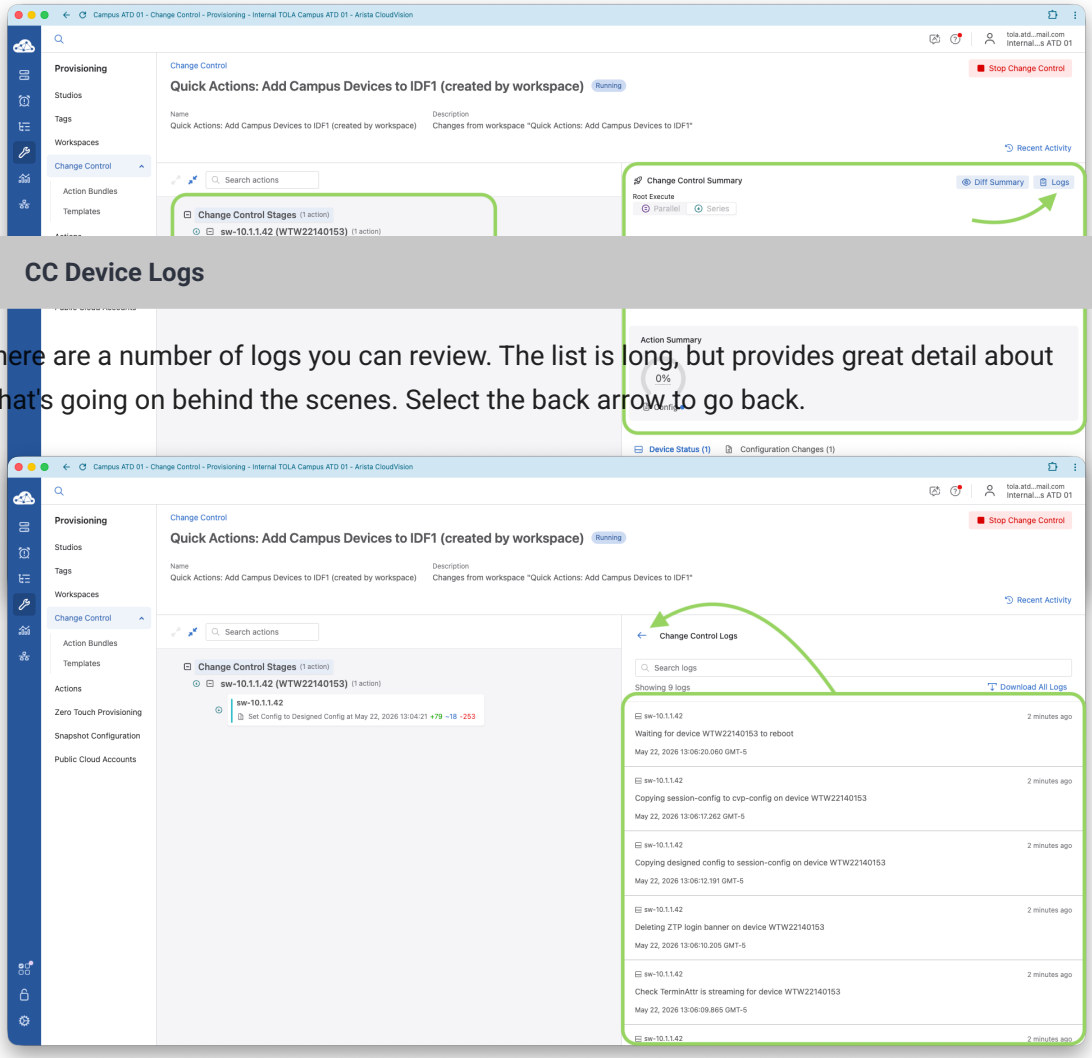
So what is the device doing?! There are a number of things the device goes through during a ZTP onboard:

a. The device configuration will be applied to the device.

b. The switch will reboot to bring the device out of ZTP mode (this will take about 10 minutes).

c. We did not upgrade the device, but if we had, the software upgrade would then take place and reboot once more.

d. Select `Logs` to view progress while the switch is rebooting.

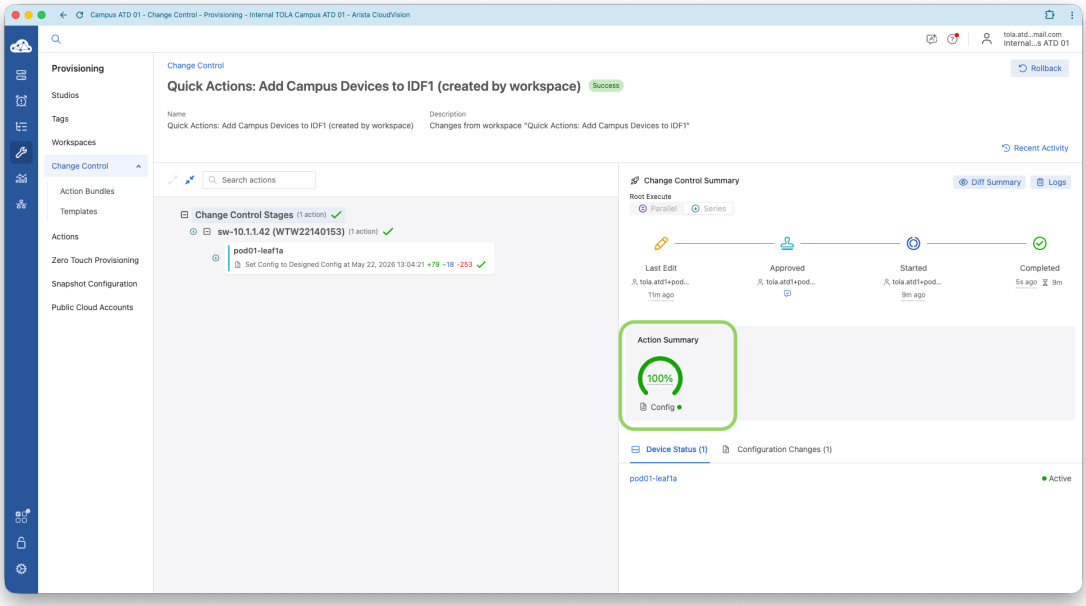


CC Device Logs

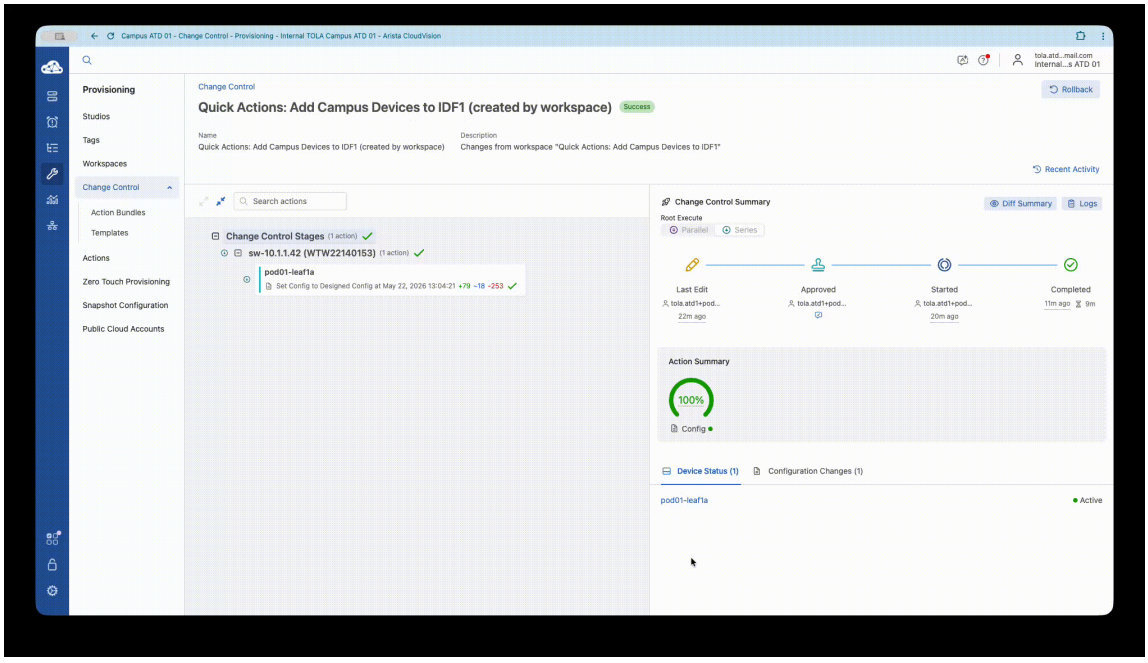
There are a number of logs you can review. The list is long, but provides great detail about what's going on behind the scenes. Select the back arrow to go back.

CC Finish

When the switch is done rebooting and streaming to CloudVision you should see 100%.



4. You can now navigate back to the `Network Hierarchy` tab on the left and see your device in the IDF with its new hostname!



5. That's it! Your new campus switch went from out-of-the-box ZTP mode to a configured member of the Campus Fabric. We're going to explore further changes to this switch in the next lab!

🎉 CONGRATS! You have completed this lab! 🎉

LET'S GO TO THE NEXT LAB!

2026-06-15